

Saketh Velidimalla

New Jersey, USA

📞 623-254-8511 • ✉️ sakethv7@gmail.com • 🔗 LinkedIn · 📁 GitHub · 📁 Portfolio

Applied AI systems and data engineer with production ownership of evaluation, observability, analytics, and governed data layers for an enterprise RAG platform serving approximately 140,000 users. Builds measurable quality infrastructure, turns fragmented operational evidence into reliable data products, and translates findings into engineering and business decisions.

Core Competencies

- **Applied AI Systems:** RAG pipelines, LLM integration, retrieval and vector search, FastAPI, Docker, AWS Bedrock, LLM classification
- **Evaluation & Observability:** LLM-as-a-Judge, regression testing, Arize Phoenix, trace analysis, hallucination and retrieval-quality analysis, red teaming
- **Data Products & Engineering:** Python, SQL, PySpark, Airflow DAGs, DynamoDB, Databricks, Delta/Parquet, AWS S3/EMR/Redshift, schema validation
- **Enterprise Analytics & Governance:** Power BI/DAX, semantic models, ServiceNow analytics, metric lineage, PII classification, retention guidance, runbooks

Professional Experience

Johnson & Johnson

Data Scientist

New Brunswick, NJ

Feb 2025–Present

- Owned evaluation, analytics, reporting, and observability for a production RAG system serving approximately 140,000 internal users across enterprise Q&A, case management, and content governance.
- Built a three-layer LLM-as-a-Judge pipeline using rules, AWS Bedrock / Llama 3.2 categorization, and keyword fallback to evaluate thousands of queries daily and surface retrieval, taxonomy, and data-quality failure modes.
- Built ground-truth regression tests with LLM-as-a-Judge similarity scoring and cross-run tracking to detect behavioral changes before deployment; deployed Arize Phoenix to investigate hallucinations, latency anomalies, and retrieval degradation.
- Designed a multi-table Power BI semantic model and KPIs for containment, query response rate, repeat users, satisfaction, escalation, and operational trends for weekly leadership reporting.
- Integrated row-level assistant interaction evidence with ServiceNow cases, external forms, and live-agent intake to distinguish same-day friction from delayed unresolved needs.
- Built a DynamoDB-to-categorized-report pipeline for knowledge-base coverage gaps and an end-to-end metrics pipeline with schema design, ingestion code, and Parquet validation into Delta Lake for Databricks reporting.
- Authored metric catalogs, lineage notes, PII-classification inputs, retention guidance, and runbooks; translated data gaps and routing deficiencies into schema and engineering requirements.

Hewlett Packard Inc.

Data Science Engineer

Spring, TX

Jul 2024–Nov 2024

- Built Airflow DAGs and PySpark transformations from S3 PC telemetry into Redshift views and Power BI reporting across hardware, software, power, sensor, and BIOS data.
- Optimized distributed PySpark processing on AWS EMR, reducing runtime by approximately 25% through query optimization and repartitioning.
- Developed and validated SQL transformations for KPI extraction and schema checks, producing reliable telemetry reporting for hardware, software, and chip-design teams.

ECrent Worldwide Company

Machine Learning Engineer

Remote

Jul 2021–Jun 2022

- Built and deployed a content-based recommendation system using structured real-estate listing attributes and text descriptions to generate personalized suggestions.
- Implemented Word2Vec semantic-similarity matching, combining structured and unstructured features to improve relevance and cold-start behavior; integrated top-N retrieval into a Flask backend.

Selected Systems Work

- **SakethWiki:** Local RAG reliability system with curated ingestion, review-gated amendments, telemetry, trace review, evaluation replay, regression gates, and task-based model routing.
- **FactoryMind:** Early sensor-to-human decision chain using ESP32 telemetry, MQTT, a FastAPI gateway, and SQLite to surface machine-health signals for human review.

Education & Development

Arizona State University

MS Information Technology (Data Science)

Phoenix, AZ

Aug 2022–May 2024

Completed: Claude Code in Action, Claude 101, Building with Claude API, Agentic AI specialization (Andrew Ng / DeepLearning.AI)

AI tooling: Codex, Claude Code **In progress:** LangGraph Fundamentals, LLM Engineering